=== UNIT 3: TRANSFORMERS, DC MOTORS & INDUCTION MACHINES ===

>> 3.1 Single-Phase Transformers: Principle and EMF Equation

A transformer is a static electromagnetic device that transfers electrical energy between two or more circuits via mutual induction without changing frequency.
* Working Principle: Alternating voltage V1 applied to primary winding of N1 turns produces an alternating magnetic flux Phi(t) in the laminated ferromagnetic core. This core flux links the secondary winding of N2 turns, inducing an alternating EMF E2 according to Faraday's Law.
* Transformer EMF Equation: Let Phi(t) = Phi_m * sin(omega * t).
e1(t) = - N1 * d(Phi)/dt = - N1 * omega * Phi_m * cos(omega * t) = N1 * omega * Phi_m * sin(omega * t - pi/2).
Peak induced EMF: E_{m1} = N1 * omega * Phi_m = 2 * pi * f * N1 * Phi_m.
RMS Induced EMF in Primary: E1 = E_{m1} / sqrt(2) = (2 * pi / sqrt(2)) * f * N1 * Phi_m = 4.44 * f * N1 * Phi_m.
RMS Induced EMF in Secondary: E2 = 4.44 * f * N2 * Phi_m.
* Transformation Ratio (K): K = E2 / E1 = N2 / N1 = I1 / I2.

>> 3.2 Transformer Losses, Efficiency and Voltage Regulation

* Core (Iron) Losses (P_i): Constant losses occurring in the magnetic core, independent of load current:
1. Hysteresis Loss (P_h): Energy dissipated per magnetization cycle: P_h = eta_Steinmetz * (B_m)^{1.6} * f * Volume.
2. Eddy Current Loss (P_e): Circulating currents induced in core iron: P_e = K_e * (B_m)^2 * f^2 * t^2 * Volume (minimized by thin, varnished silicon-steel laminations of thickness t ~ 0.35 mm).
* Copper (Ohmic) Losses (P_cu): Variable I^2*R losses in windings: P_cu = I1^2 * R1 + I2^2 * R2 = I2^2 * R_{02}.
* Condition for Maximum Efficiency: Maximum efficiency occurs when variable copper losses equal constant iron losses: P_cu = P_i. Load current at maximum efficiency: I_{2, max} = I_{2, rated} * sqrt(P_i / P_{cu, rated}).
* Voltage Regulation: Percentage change in secondary terminal voltage from no-load to full-load: %VR = [(V_{2, no-load} - V_{2, full-load}) / V_{2, rated}] * 100 = [(I2 * R_{02} * cos(phi) +/- I2 * X_{02} * sin(phi)) / V2] * 100 (+ for lagging PF, - for leading PF).

>> 3.3 Direct Current (DC) Machines: Motors and Generators

* DC Motor Principle: Operates on Lorentz force: a current-carrying conductor placed in a magnetic field experiences mechanical force F = B * I * L * sin(theta) directed

according to Fleming's Left-Hand Rule.
* Back EMF (E_b): As the armature rotates, its conductors cut magnetic flux, inducing a counter-EMF opposing supply voltage: $E_b = (P * Phi * Z * N) / (60 * A)$, where P = poles, Phi = flux/pole, Z = total armature conductors, N = RPM, A = parallel paths (A = 2 for wave winding, A = P for lap winding).
* Armature Voltage Equation: $V = E_b + I_a * R_a$.
* Armature Torque Equation: Electromagnetic power converted = $E_b * I_a = T_e * omega_m = T_e * (2*pi*N / 60)$.
$T_e = [(P * Z) / (2 * pi * A)] * Phi * I_a = 0.159 * (P * Z / A) * Phi * I_a$ (Torque is directly proportional to flux and armature current: T_e propto Phi * I_a).

>> 3.4 Three-Phase Induction Motors

* Rotating Magnetic Field (RMF): Balanced 3-phase currents flowing through spatially displaced stator windings (120 degrees apart) produce a magnetic field of constant magnitude (1.5 * Phi_m) rotating at Synchronous Speed $N_s = 120 * f / P$.
* Rotor Slip (s): Difference between synchronous stator speed and actual rotor speed N_r: $s = (N_s - N_r) / N_s$.
* Rotor Current and Torque: Stator RMF cuts stationary rotor bars, inducing rotor currents. Induced currents interact with RMF to produce torque pulling the rotor in the direction of the rotating field. A motor can never run at synchronous speed (if $N_r = N_s$, slip = 0, no cutting of flux, torque collapses to zero).

[SOLVED PROBLEM 1: Transformer EMF, Turns and Flux Density]

Problem: A single-phase 2,200 / 220 V, 50 Hz step-down transformer has a net core cross-sectional area of 150 cm^2 (0.015 m^2). If the maximum flux density in the core is B_m = 1.2 Tesla, calculate: (1) Maximum magnetic flux Phi_m, (2) Number of primary turns N1, (3) Number of secondary turns N2, (4) Primary and secondary full-load currents for a 33 kVA rating.
Solution Step 1: Compute Maximum Core Flux Phi_m:
$Phi_m = B_m * A = 1.2 T * 0.015 m^2 = 0.018$ Webers (18 mWb).
Solution Step 2: Calculate Number of Primary Turns N1:
$E1 = 4.44 * f * N1 * Phi_m => N1 = E1 / (4.44 * f * Phi_m)$
$N1 = 2200 / (4.44 * 50 * 0.018) = 2200 / 3.996 = 550.55 => 551$ turns.
Solution Step 3: Calculate Number of Secondary Turns N2:
Transformation ratio $K = E2 / E1 = 220 / 2200 = 0.10$.
$N2 = K * N1 = 0.10 * 551 = 55.1 => 55$ turns.
Solution Step 4: Compute Full-Load Currents for 33 kVA Rating:
Rated Primary Current $I1 = Rated kVA * 1000 / V1 = 33,000 / 2200 = 15.0$ Amperes.
Rated Secondary Current $I2 = Rated kVA * 1000 / V2 = 33,000 / 220 = 150.0$ Amperes.

[SOLVED PROBLEM 2: Transformer Maximum Efficiency and Load Optimization]

Problem: A 25 kVA single-phase transformer has a constant core iron loss of 250 W and a full-load copper loss of 400 W. Calculate: (1) Efficiency at full load with 0.8 power factor lagging, (2) The kVA load at which maximum efficiency occurs, (3) Value of maximum efficiency at unity power factor.

Solution Step 1: Full-Load Efficiency at 0.8 PF Lagging:

Output Power = Rated kVA * PF = 25 * 0.8 = 20.0 kW = 20,000 W.

Total Losses = Iron Loss + Full-Load Copper Loss = 250 W + 400 W = 650 W.

Input Power = Output + Losses = 20,000 + 650 = 20,650 W.

Efficiency eta = (Output / Input) * 100 = (20,000 / 20,650) * 100 = 96.85%.

Solution Step 2: kVA Load for Maximum Efficiency:

Condition: Variable Copper Loss = Constant Iron Loss.

kVA_max = Rated kVA * sqrt(P_iron / P_cu_FL) = 25 * sqrt(250 / 400) = 25 * sqrt(0.625)

kVA_max = 25 * 0.7906 = 19.76 kVA.

Solution Step 3: Maximum Efficiency at Unity Power Factor (PF = 1.0):

Output Power at max efficiency = 19.76 kVA * 1.0 = 19.76 kW = 19,760 W.

Total Losses at max efficiency = Iron Loss + Copper Loss = 250 W + 250 W = 500 W.

Input Power = 19,760 + 500 = 20,260 W.

eta_max = (19,760 / 20,260) * 100 = 97.53%.

[SOLVED PROBLEM 3: DC Shunt Motor Back EMF and Armature Torque]

Problem: A 230V DC shunt motor has an armature resistance R_a = 0.5 Omega and field resistance R_sh = 115 Omega. At full load, it draws a total line current of 22A and runs at 1,200 RPM. Calculate: (1) Shunt field current I_sh, (2) Armature current I_a, (3) Back EMF E_b, (4) Developed electromagnetic torque T_e.

Solution Step 1: Calculate Field Current I_sh:

I_sh = V / R_sh = 230 V / 115 Omega = 2.0 Amperes.

Solution Step 2: Calculate Armature Current I_a:

I_a = I_line - I_sh = 22.0 A - 2.0 A = 20.0 Amperes.

Solution Step 3: Compute Back EMF E_b:

E_b = V - I_a * R_a = 230 - (20.0 * 0.5) = 230 - 10.0 = 220.0 Volts.

Solution Step 4: Compute Electromagnetic Torque T_e:

Mechanical Power converted P_mech = E_b * I_a = 220 V * 20 A = 4,400 Watts.

Angular Velocity omega_m = 2 * pi * N / 60 = 2 * pi * 1200 / 60 = 40 * pi = 125.66 rad/sec.

Torque T_e = P_mech / omega_m = 4400 / 125.66 = 35.01 N*m (Newton-meters).

[SOLVED PROBLEM 4: 3-Phase Induction Motor Slip and Rotor Parameters]

Problem: A 4-pole, 50 Hz, 3-phase induction motor runs at a rotor speed of 1,440 RPM at full load. Calculate: (1) Synchronous speed N_s, (2) Slip s, (3) Percentage slip %s, (4) Frequency of rotor induced currents f_r.

Solution Step 1: Compute Synchronous Speed N_s:

N_s = 120 * f / P = (120 * 50) / 4 = 6,000 / 4 = 1,500 RPM.

Solution Step 2: Calculate Fractional Slip s:

s = (N_s - N_r) / N_s = (1500 - 1440) / 1500 = 60 / 1500 = 0.04 (4.0%).

Solution Step 3: Calculate Rotor Current Frequency f_r:

f_r = s * f = 0.04 * 50 Hz = 2.0 Hz.

Conclusion: Under operating conditions, rotor currents oscillate at only 2 Hz, minimizing rotor iron core losses.

>> 3.5 University Review & Examination Questions

* Q1: Derive the condition for maximum efficiency of a transformer from first principles.

Answer: Let transformer rating be S (VA), output power P_out = V2 * I2 * cos(phi). Total losses are Iron Loss (P_i, constant) and Copper Loss (P_cu = I2^2 * R_02, variable). Efficiency is eta = P_out / (P_out + Losses) = [V2 * I2 * cos(phi)] / [V2 * I2 * cos(phi) + P_i + I2^2 * R_02]. Divide numerator and denominator by I2: eta = [V2 * cos(phi)] / [V2 * cos(phi) + P_i / I2 + I2 * R_02]. For maximum efficiency with respect to load current I2, the denominator must be minimized. Differentiating denominator with respect to I2 and setting to zero: d/dI2 [V2*cos(phi) + P_i / I2 + I2 * R_02] = - P_i / (I2)^2 + R_02 = 0 => P_i = (I2)^2 * R_02 = P_cu. Therefore, maximum efficiency occurs when variable copper loss equals constant core iron loss.

* Q2: Explain why a 3-phase induction motor is self-starting whereas a single-phase induction motor is not.

Answer: When 3-phase balanced currents flow through 3-phase stator windings spaced 120 degrees apart, they produce a true Rotating Magnetic Field (RMF) of constant magnitude 1.5*Phi_m that rotates at synchronous speed N_s. This RMF cuts stationary rotor conductors, inducing currents that create a unidirectional starting torque. In contrast, a single-phase stator winding produces a pulsating stationary magnetic field that alternates along a single spatial axis without rotation. By Double Revolving Field Theory, a pulsating field decomposes into two equal fields rotating in OPPOSITE directions at speed N_s. At standstill (N = 0), both fields exert equal and opposite torques; net starting torque is identically zero. Hence, single-phase motors require auxiliary starting windings or split-phase capacitors.

>> 3.6 Key Takeaways and Revision Points

* Transformer EMF equation: E = 4.44 * f * N * Phi_m.
* Maximum transformer efficiency occurs when copper loss equals core iron loss (P_cu = P_i).

* DC motor back EMF: $E_b = V - I_a * R_a$; electromagnetic torque: T_e propto Phi * I_a.
* 3-phase induction motor synchronous speed $N_s = 120*f / P$; rotor frequency $f_r = s*f$.